

Penetration Testing Methodologies

- Penetration Testing Methodology is a step-by-step process used by ethical hackers to find security weaknesses in a computer system, network, or web application.
 - It helps organizations identify vulnerabilities before real hackers attack the system.
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Phases of Penetration Testing Methodology

1. Planning and Reconnaissance

In this phase, the tester collects information about the target system.

Activities:

- Identifying target systems
 - Collecting IP addresses
 - Gathering domain information
 - Understanding network structure
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2. Scanning

In this phase, the tester scans the target system to identify open ports, services, and vulnerabilities.

Activities:	Tools Used:
● Port scanning ● Network scanning ● Vulnerability scanning	● Nmap ● Nessus

3. Gaining Access

In this phase, the tester tries to enter the system using discovered vulnerabilities.

Activities:

- Password attacks
- SQL injection
- Exploiting software vulnerabilities

Example: Using weak passwords to log into a server.

4. Maintaining Access

The tester checks whether an attacker can stay inside the system for a long time.

Activities:

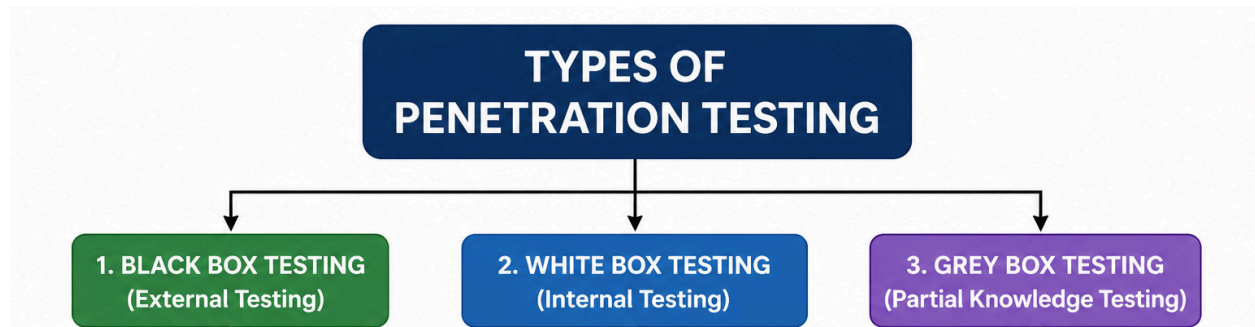
- Installing backdoors
 - Creating hidden accounts
 - Escalating privileges
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5. Analysis and Reporting

After testing, the tester prepares a report of all vulnerabilities found.

Report Includes:

- Vulnerabilities identified
- Risk level
- Solutions and recommendations



Advantages of Penetration Testing

1. Identifies security weaknesses
2. Prevents cyber attacks
3. Protects sensitive information
4. Improves network security

TCP/IP Model

- TCP/IP stands for **Transmission Control Protocol / Internet Protocol**.
 - The TCP/IP model is a networking model used for communication over the Internet and computer networks,
 - Each layer performs a specific function in the communication process.
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Layers of TCP/IP Model

1. Application Layer
 2. Transport Layer
 3. Internet Layer
 4. Network Access (Link) Layer
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1. Application Layer

- The Application Layer is the topmost layer of the TCP/IP model.
- It provides services directly to users and applications.

Functions:

- Supports web browsing and email communication
- Provides file transfer services

Example Protocols

- HTTP / HTTPS – Web browsing
- FTP – File transfer
- SMTP – Sending email
- POP3 / IMAP – Receiving email

2. Transport Layer

- The Transport Layer provides end-to-end communication between devices.

Functions:

- Breaks data into segments
- Ensures reliable data delivery
- Performs error checking

Main Protocols

TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
1. Reliable and connection-oriented 2. Ensures correct delivery of data	1. Faster but unreliable 2. No guarantee of delivery

3. Internet Layer

The Internet Layer handles logical addressing and routing of packets between networks.

Functions

- Assigns IP addresses
- Routes packets between networks
- Performs packet forwarding

Protocols used in Internet Layer

- IP (Internet Protocol)
 - ICMP – Error reporting
 - ARP – Finds physical address
 - IGMP – Multicast communication
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4. Network Access (Link) Layer

- This is the lowest layer of the TCP/IP model.
- It handles physical transmission of data through network media.

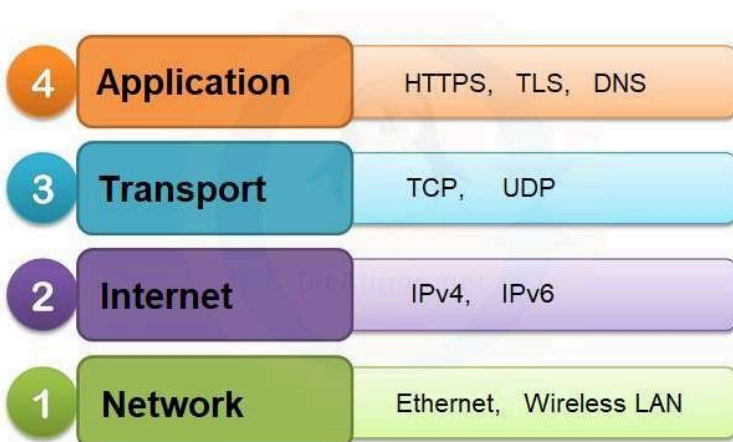
Functions

- Converts packets into frames
- Uses MAC addresses

Examples

- Ethernet
 - Wi-Fi
 - Optical Fiber
 - LAN technologies
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Simple Diagram of TCP/IP Model



Advantages of TCP/IP Model

1. Open standard and widely used
2. Supports large networks
3. Works with different hardware devices
4. Provides reliable communication

Network and Computer Attacks

- A Network and Computer Attack is an attempt made by hackers or attackers to damage, steal, access, or disrupt computer systems, networks, or data without permission.
 - These attacks can affect personal computers, servers, websites, and entire networks.
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Types of Network and Computer Attacks (Any 5-6)

1. Malware Attack

Malware means malicious software designed to damage or control a computer system.

Types of Malware

- Virus
- Worm
- Trojan Horse
- Ransomware
- Spyware

Effects

- Data loss
 - Slow system performance
 - Unauthorized access
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2. Phishing Attack

Phishing is a fake email or website attack used to steal sensitive information.

Purpose

- Steal passwords
- Steal bank details
- Steal personal information

Example: A fake bank email asking for login details.

3. Denial of Service (DoS) Attack

In a DoS attack, the attacker sends huge traffic to a server to make it unavailable.

Effects

- Website becomes slow
 - Server crashes
 - Services stop working
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4. Password Attack

Attackers try to guess or steal passwords to access systems.

Methods

- Brute force attack
- Dictionary attack
- Keylogging

5. Man-in-the-Middle (MITM) Attack

In this attack, the attacker secretly intercepts communication between two users.

Effects

- Data theft
 - Message modification
 - Privacy loss
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6. SQL Injection Attack

Attackers insert malicious SQL commands into websites to access databases.

Effects

- Database theft
 - Data modification
 - Unauthorized access
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7. Spoofing Attack

Spoofing means pretending to be another user or device.

Types

- IP Spoofing
- Email Spoofing
- Website Spoofing

8. Sniffing Attack

Sniffing is the process of capturing network data packets.

Purpose

- Steal usernames and passwords
 - Monitor network traffic
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9. Insider Attack

An insider attack is performed by a person inside the organization.

Example: An employee stealing company data.

Protection Against Attacks

1. Use strong passwords
 2. Install antivirus software
 3. Update software regularly
 4. Use firewalls
 5. Avoid suspicious emails and links
 6. Enable encryption
 7. Use intrusion detection systems
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Advantages of Studying Attacks

1. Helps improve security
2. Prevents cyber crimes
3. Protects sensitive information
4. Helps ethical hackers identify vulnerabilities

Malware

- Malware stands for **Malicious Software**.
 - Hackers use malware to attack computers, steal information, and control systems.
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Types of Malware

1. Virus

A Virus is a malicious program that attaches itself to files or programs.

Features

- Corrupts files and programs
- Slows down the computer

Example: Infected USB files.

2. Worm

A Worm is a self-replicating malware that spreads automatically through networks.

Features

- Does not need user action
 - Consumes network bandwidth
 - Spreads very quickly
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3. Trojan Horse

A Trojan Horse is malware disguised as useful software.

Features

- Creates backdoors for attackers
- Steals sensitive information

Example: Fake game or software download.

4. Ransomware

Ransomware locks or encrypts files and demands money to restore them.

Features

- Encrypts files
 - Displays ransom message
 - Causes data loss
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5. Spyware

Spyware secretly monitors user activities.

Features

- Tracks browsing history
 - Steals passwords
 - Sends information to attackers
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6. Adware

Adware automatically displays unwanted advertisements.

Features

- Shows popup ads
- Slows system performance
- Tracks user activity

Effects of Malware	Protection Against Malware
• Data theft • System crash • Financial loss • Slow computer performance • Unauthorized access • Loss of privacy	• Install antivirus software • Update operating systems regularly • Avoid unknown email attachments • Download software from trusted websites • Use firewalls • Scan USB devices before use • Use strong passwords

Advantages of Malware Analysis

1. Helps identify security weaknesses
2. Improves cybersecurity systems
3. Helps develop antivirus software
4. Prevents future cyber attacks

